

CERTIFICATE

This is to certify that Mr./Ms. _____

of _____ Class, Roll No. _____

Exam No. _____ has satisfactorily completed his / her term

work in _____ for

the term ending _____ in 20____ / 20____.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANGA – 388 421

Date :

Sign of the Faculty

Head of the Department

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

M. S. Patel Department of Civil Engineering

ADVANCED STRUCTURAL ANALYSIS (CL 741)

INDEX

Sr. No.	Date	Title of Tutorial	Page No.	Marks /Grade	Date of Assessment	Sign of Faculty
1		BEAM				
2		PLANE TRUSS				
3		PLANE FRAME				
4		GRID				
5		SPACE TRUSS AND SPACE FRAME				
6		SYMMETRY AND ELASTIC SUPPORT				

TUTORIAL NO. 1

DIRECT STIFFNESS METHOD

BEAM

Q-1 Analyze the two span continuous beam shown in figure 1.1.

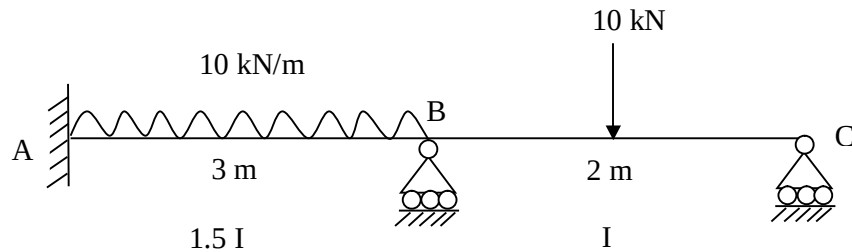


Fig: 1.1

Q-2 Calculate all unknown reactions for the continuous beam shown in Figure 1.2 using direct stiffness method.

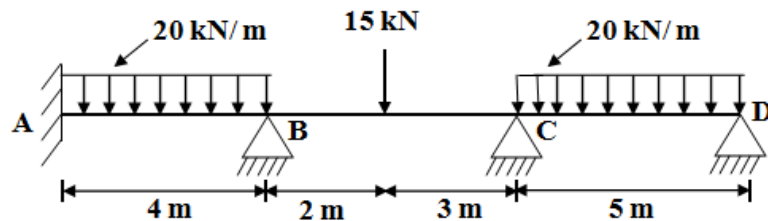


Fig: 1.2

Q-3 A continuous beam ABCD is carrying distributed load of 5 kN/m as shown in figure 1.3. Compute reactions due to following support settlements. Take EI as constant.

Support B - 0.005m vertically downward

Support C - 0.010m vertically downward

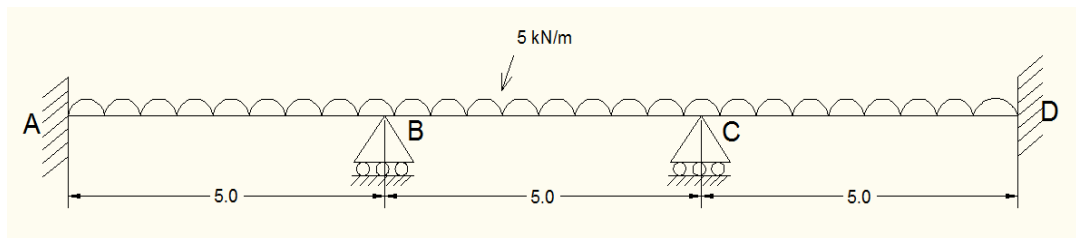


Fig: 1.3

Q-4 Calculate all unknown reactions for the continuous beam shown in Figure 1.4 using direct stiffness method. Support B sinks by 5 mm and support C sinks by 10 mm. Take E as 2 x

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

M. S. Patel Department of Civil Engineering

10^5 N/mm^2 and I as $4 \times 10^{-4} \text{ m}^4$. Consider I as constant throughout the span.

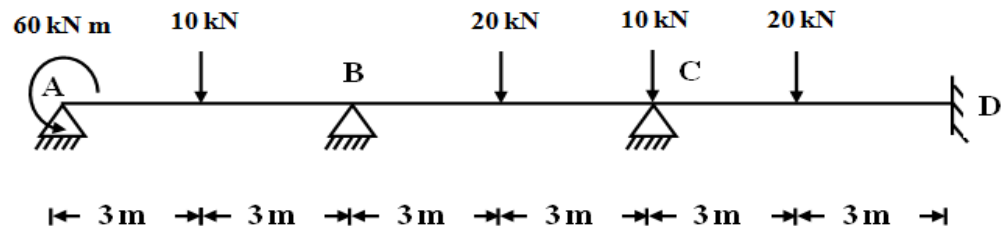
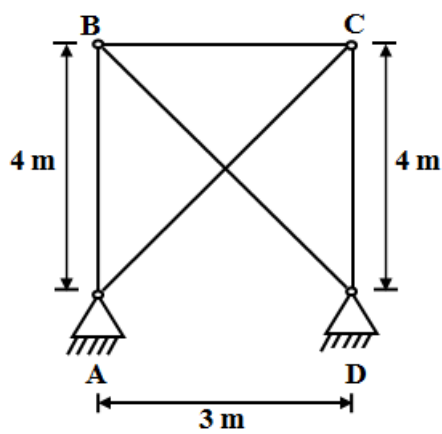
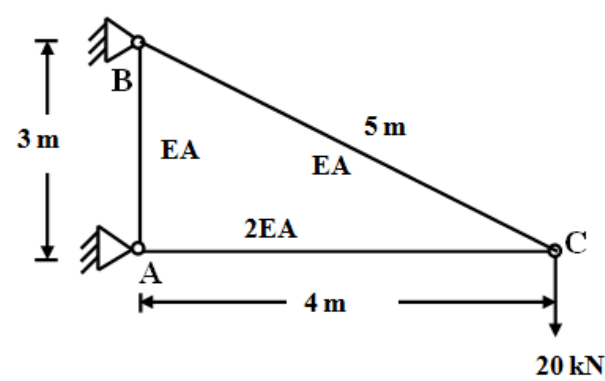


Fig: 1.4

Marks Obtained:	Signature of Faculty:	Date:
-----------------	-----------------------	-------

TUTORIAL NO. 2
DIRECT STIFFNESS METHOD
PLANE TRUSS

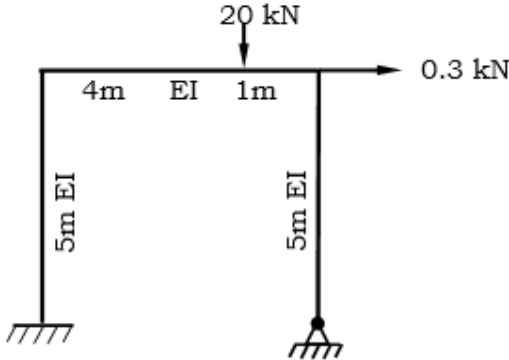
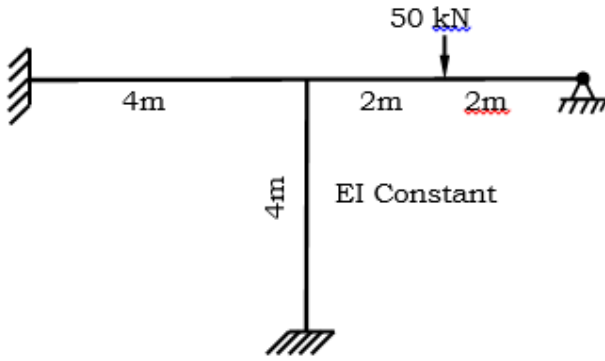
Q-1	Analyse the plane truss shown in Figure 2.1 if support A sinks by 3 mm. Take area for member AB, BC & CD as 25 cm^2 and for member BD and AC as 20 cm^2. E as $2 \times 10^5 \text{ N/mm}^2$.  Figure: 2.1
Q-2	Analyse the plane truss shown in Figure 2.2 using stiffness method member approach. Support A moves in left by 3 mm. Consider $E = 2 \times 10^5 \text{ N/mm}^2$ and $A = 1000 \text{ mm}^2$ for all members. Also draw axial force diagram.  Figure: 2.2

Marks Obtained:	Signature of Faculty:	Date:
-----------------	-----------------------	-------

Advanced Structural Analysis (CL741)

Date:

TUTORIAL NO. 3
DIRECT STIFFNESS METHOD
PLANE FRAME

Q-1	Find unknown reactions for the below plane frame structure by using Stiffness Method Member Approach. Consider $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 2 \times 10^{-3} \text{ m}^4$. Refer Figure 3.1.  Figure: 3.1
Q-2	Analyse the plane frame shown in figure 3.2 using Stiffness Method Member Approach. Consider $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 2 \times 10^{-3} \text{ m}^4$.  Fig.: 3.2

Marks Obtained:	Signature of Faculty:	Date:
-----------------	-----------------------	-------

TUTORIAL NO. 4
DIRECT STIFFNESS METHOD
GRID

Q-1 Find unknown reactions for the below grid structure by using Stiffness Method Member Approach. Consider $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 2 \times 10^{-3} \text{ m}^4$, $GJ = 0.5 EI$. Refer **Figure 4.1**.

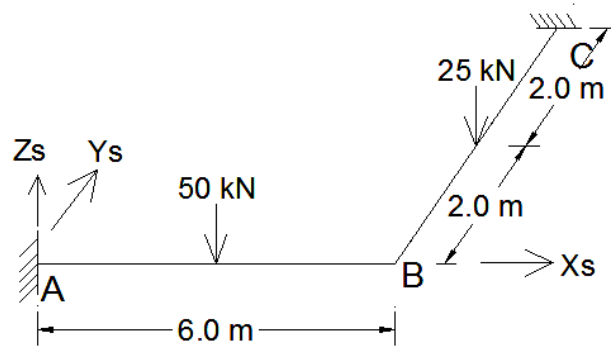


Figure: 4.1

Q-2 Analyse the grid as shown in **figure 4.2**. Take $GJ = 0.8 EI$, $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 0.001 \text{ m}^4$

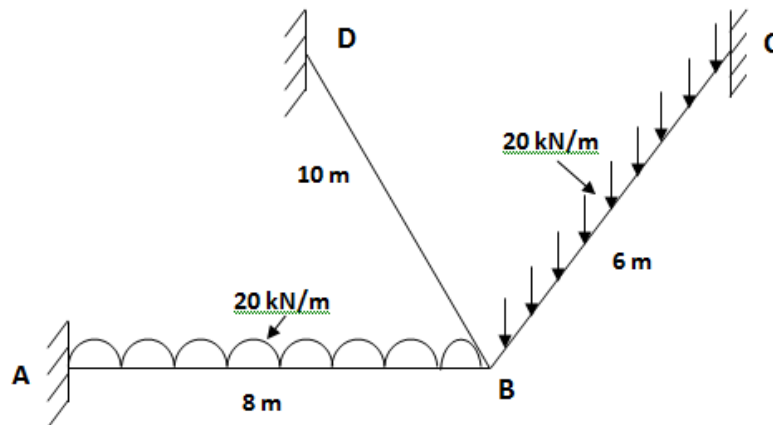


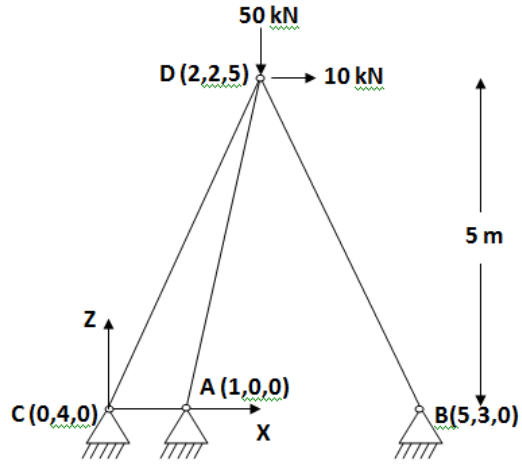
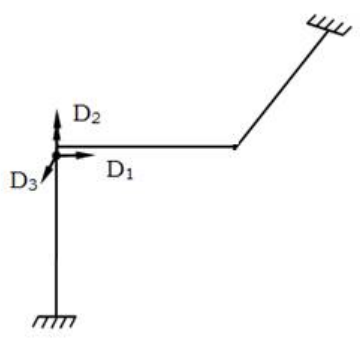
Fig.: 4.2

Marks Obtained:	Signature of Faculty:	Date:
-----------------	-----------------------	-------

Advanced Structural Analysis (CL741)

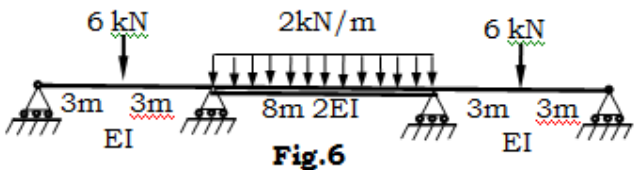
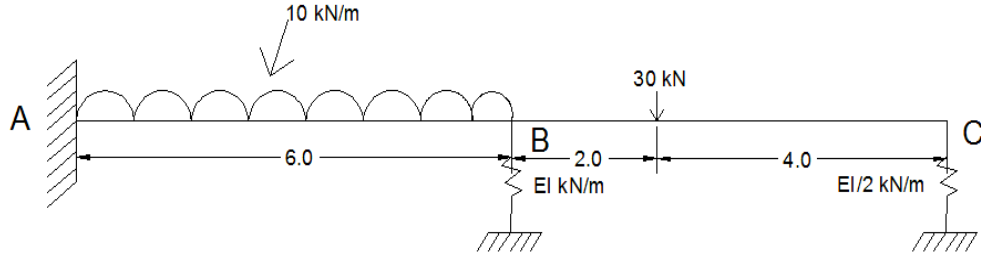
Date:

TUTORIAL NO. 5
DIRECT STIFFNESS METHOD
Space Truss & Space Frame

Q-1	Write the space truss shown in figure 5.3. Take $E = 200 \text{ kN/mm}^2$ and $A = 1000 \text{ mm}^2$ for all members.  Fig: 5.1
Q-2	Write the member stiffness matrix in member axes and rotation matrix for the space frame member.
Q-3	Develop Member stiffness matrix for the given Space frame for the shown in free displacements. Consider L, EI, EA and GJ as constant for all members. Also consider structural axes as reference axes.  Fig. 5.2

Marks Obtained:	Signature of Faculty:	Date:
-----------------	-----------------------	-------

TUTORIAL NO. 6
DIRECT STIFFNESS METHOD
Symmetry and Elastic Support

Q-1	Analyse the given 3-span continuous beam by using Stiffness member approach with concept of symmetry.  Fig.6
Q-2	Analyse the beam shown in figure 5.4. by stiffness method. Take EI as constant.  Fig.: 6.2

Marks Obtained:	Signature of Faculty:	Date:
-------------------------------	-------------------------------------	---------------------